Journal of the Brazilian Society of Mechanical Sciences and Engineering (2026) 48:179
https://doi.org/10.1007/s40430-025-06206-4

TECHNICAL PAPER

Development and optimization of Hammerstein-Wiener system identification for modelling of super elastic shape memory alloy wire

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Received: 26 February 2025 / Accepted: 24 November 2025

Abstract

Shape memory alloys are valuable materials for a distinct engineering applications such as aerospace, robotics, medical etc., due to effect of their superelasticity and shape memory. The superelastic shape memory alloy wire (SESMA) was selected with a composition of 54.3% Ni and 45.7% Ti, which also has a diameter of 0.6 mm and a length of 30 mm. A thermomechanical machine was used to perform cyclic testing up to a 6% strain level at a constant temperature (23 °C). The experiments were performed, and data were obtained on the stress-strain. The modelling of cyclic behaviour is developed by using a Hammerstein-Wiener technique. Moreover, Hammerstein-Wiener (HW) models were obtained using the second cycle and validated through fifth cycle with experimental data. The results showed that the model's hysteresis loop was reasonably close to the experimental data for all frequencies. Further, the austenite modulus and area of the stress-strain hysteresis loop were found to have a good correlation with a maximum error of 1.29%. These results further suggest that Hammerstein-Wiener (HW) system identification is very useful for modeling the SESMA behaviour.

Keywords Super elastic shape memory alloys · Hysterisis loop · System identification · Wavelet network · Levenberg-Marquardt algorithm

1 Introduction

Shape memory alloys (SMA) are a type of engineered material that returns to its original shape when it is heated/loaded. SMA has two phases: the martensite phase, which has a low yield strength and is easily deformed below the transition temperature [1]. SMA's crystal structures undergo a phase transformation at a temperature above the transition temperature and revert to the parent phase or austenite phase [2, 3]. Temperature-induced transformations in SMA involve phase changes between martensite and austenite, enabling the shape memory effect. Such a type of material has found a wide range of research in the aerospace and automotive industries. Changes from the austenite to the martensite phase, which are caused by the application of external loading, manifest as stress-induced transformations. A considerable amount of recoverable strain (~ 8%) is generated in the material during such a process, and the material exhibits super elasticity [4–6]. Due to their damping nature, super elastic shape memory alloys (SESMA) are employed in various aerospace and seismic applications [7, 8]. Modelling of SMA is essential for a comprehensive understanding of alloys in aerospace and biomedical applications. Different studies initially focus on the small diameter of SMA wire, including this article, for application-specific purposes. Limited studies have been conducted on SMA wires with a large diameter. In structural applications, it is important to handle larger diameters of SMA wire, which are theoretically advantageous due to their increased load-carrying capacity and exceptional hysteretic energy dissipation. However, they may also experience altered phase transformation dynamics as a result of increased cross-sectional constraint and nonuniform stress distribution. To address this lacuna, Zhang et al. [9] present a study that systematically examines

Technical Editor: Marcelo A. Savi

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Published online: 11 February 2026

3 mm diameter Ni–Ti wires under controlled mechanical loading conditions.

Macroscopic models, microscopic models, and mixed-mode models are the three broad categories into which the reported models can be categorized [10–14]. The models try to represent the behaviour of SMAs by drawing on a wide variety of theories and principles, but only a select few of these models are actually useful and applicable in aerospace and seismic contexts. This is due to the fact that some SMA constitutive models are too complex and difficult to incorporate numerically in simulations. Even after all of the parameters are precisely determined, constitutive models still need an additional tuning step in order to provide accurate response estimations. This is one of the most significant aspects while modelling SMA numerically.

On the other hand, a data-driven approach that includes neuro-fuzzy algorithms, machine learning (ML) approaches, and system identification gives an elegant solution that circumvents these issues that have arisen in constitutive modelling. Ozbulut et al. [15] demonstrated a neuro-fuzzy model of NiTi SMA wires for seismic applications. Using experimental input-output data pairs, an adaptive neuro-fuzzy inference system (ANFIS) of SMA is developed. The ANFIS-trained model is validated using an experimental data set that is not used during training. The model was used to predict superelastic SMA behaviour at different temperatures and loading rates while being simple enough for numerical computations. Tymoshchuk et al. [16] developed a machine learning process that can predict the loading frequency of nickel-titanium shape memory alloys from stress, strain, and cycle data. Among several models tested, a neural network delivered the best accuracy. The findings could help improve real-time monitoring and maintenance of devices that rely on these smart materials. Lenzen et al. [17] demonstrate a neural network-based machine learning method that can effectively and precisely identify key material parameters of superelastic shape memory alloy (SMA) wires from cyclic tensile data. For engineering applications, this method considerably simplifies the modeling of their dynamic, strain-rate-dependent behavior.

Among various data-driven and hybrid modeling techniques like neural-network and fuzzy-based methods provide strong predictive performance, their physical interpretability remains limited. To bridge this gap, researchers have increasingly turned to block-oriented system identification frameworks such as Hammerstein–Wiener models.

The system identification method for Hammerstein, Wiener, or Hammerstein-Wiener processes is a widely used methodology for modelling a nonlinear dynamical system [18–20]. In recent years, identification of Hammerstein–Wiener models has seen renewed interest. For example, Copaci et al. [21] demonstrated SMA wire as an actuator for robotic applications. Here, two stages, i.e., control-temperature and temperature position, are modelled using a Hammerstein-Wiener model. In this article results show that the model closely resembles the response of an actual actuator. Furthermore, Hammerstein models as a solution to pneumatic muscle actuator hysteresis have been demonstrated by Liu et al. [22]. This study intends to illustrate Hammerstein models coupled with a backpropagation neural network, assess their ability to capture nonlinear behaviour, and analyze their use in pneumatic muscle actuators to reduce hysteresis. It emphasizes the need for proper modeling and control in maximizing pneumatic muscle actuator performance.

Recent developments in block-oriented nonlinear system identification have further advanced the modelling of Hammerstein, Wiener, and Hammerstein–Wiener structures. Sun et al. [23] proposed a robust HW model identification method designed for highly nonlinear systems with short and noisy data sets. Brouri et al. [24] developed a method for identification of HW models with irregular input nonlinearity, demonstrating experimental strategies to extract input/output nonlinearities and the linear subsystem. A multi-model HW approach for fast and accurate modelling was presented by Chihi et al. [25] targeting diverse force profiles in actuator systems. The selection of HW structure in our work is thus anchored in this recent methodological development: HW models provide a compact yet expressive block-oriented representation for strongly nonlinear dynamic behaviour, while retaining interpretability of the linear subsystem and static input/output nonlinearities. This is particularly advantageous when modelling SESMA that exhibits large hysteretic loops, strain-rate dependence, and frequency-varying behaviour across 0.01 Hz to 30 Hz.

This paper presents the Hammerstein-Wiener method for modelling the cyclic behavior of SESMA using experimental stress-strain data. Hammerstein-Wiener (HW) models are developed, in particular, by employing the second cycle of the experimental data using 0.01 Hz, 0.1 Hz, 1 Hz, 10 Hz, 20 Hz, and 30 Hz frequencies under test. Through an iterative approach with MATLAB's system identification toolbox, the model's architecture and properties are determined. In modeling of the Hammerstein-Wiener method, piecewise-linear (PL) and wavelet-network (WN) functions are applied as nonlinear estimators for input and output channels that vary across different frequencies. The linear component of the model is kept constant for optimal fit to the experimental data.

The SESMA response under various loading frequencies is highly nonlinear and hysteretic in nature; therefore, optimization algorithms are essential for accurate and robust model performance. In this work, the optimization algorithms used for parameter estimation, including the

Gauss-Newton algorithm, which linearizes the model to minimize error, and the Levenberg-Marquardt algorithm, which combines Gauss-Newton and gradient descent for robust convergence, are discussed in detail in Sect. 4.

Further, to understand energy dissipation with respect to the volume of the wire, we have compared it with other cited works. In our experiment, a NiTi wire with a diameter of 0.6 mm and a gauge length of 30 mm was used. At a frequency of 1 Hz, the dissipation energy was 7.949 MJ/m³. In comparison, Silva et al. [26] used a NiTi micro-cable with a diameter of 0.4064 mm and a gauge length of 12.6 mm, and reported a dissipation energy of 4.22 MJ/m³ at 1 Hz. Similarly, Ramos et al. [27] used a NiTi wire with a diameter of 0.5 mm and a gauge length of 20 mm, and at 1 Hz, the dissipation energy was 15 MJ/m³.

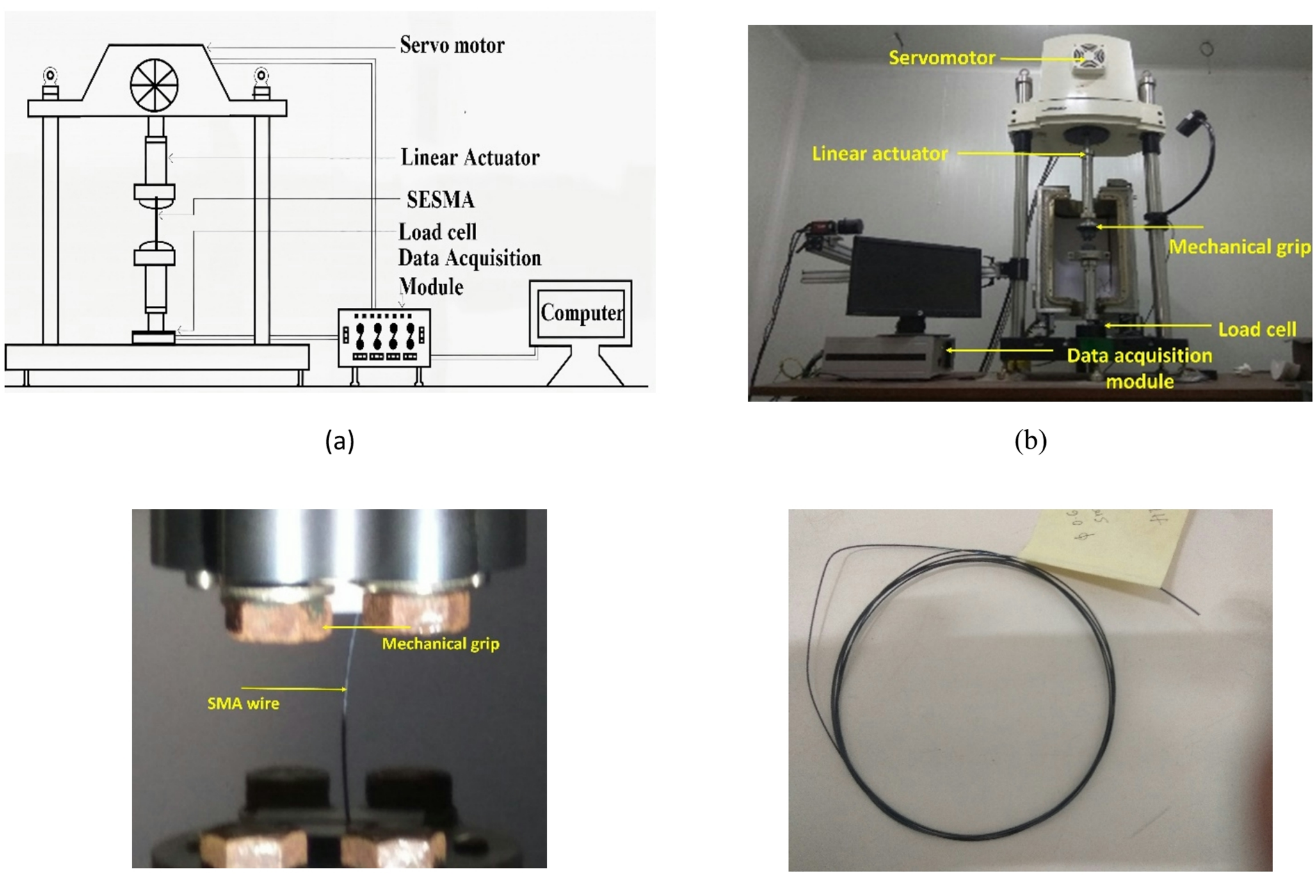
In addition, the area of the stress-strain hysteresis loop corresponding to dissipation energy (Ed) and the austenite modulus are computed from model output and compared to experimental data, where they are shown to be well correlated.

The paper's structure unfolds as follows: Sect. 2 provides a detailed exposition of the experimental setup. Section 3 delves into the intricacies of the modelling approach, elucidating the theoretical framework and methodology used. Section 4 focuses on the optimization of the Hammerstein-Wiener model, discussing the strategies and algorithms applied. Section 5 is dedicated to presenting and discussing the results. Finally, Sect. 6 encapsulates the paper with conclusive remarks, synthesizing the key findings and their implications.

2 Experimental setup

In this section, the experimental setup was developed to conduct the tensile cycle at different frequencies. A SESMA wire was studied using tension-tension cycling testing. The testing was carried out using a computerised thermomechanical fatigue testing machine (Model: Bose Electroforce 3330), as shown in Fig. 1, with the equipment set to displacement control mode.

This elongation length is defined as the cross-head movement. The experimental test setup was kept in a temperature stability chamber. The temperature of the chamber was



(c) (d)

Fig. 1 **a** Block diagram of low-cycle fatigue test experiment, **b** Picture of test setup, **c** Enlarged representation of the wire in the tension grip, **d** Sample of SMA wire

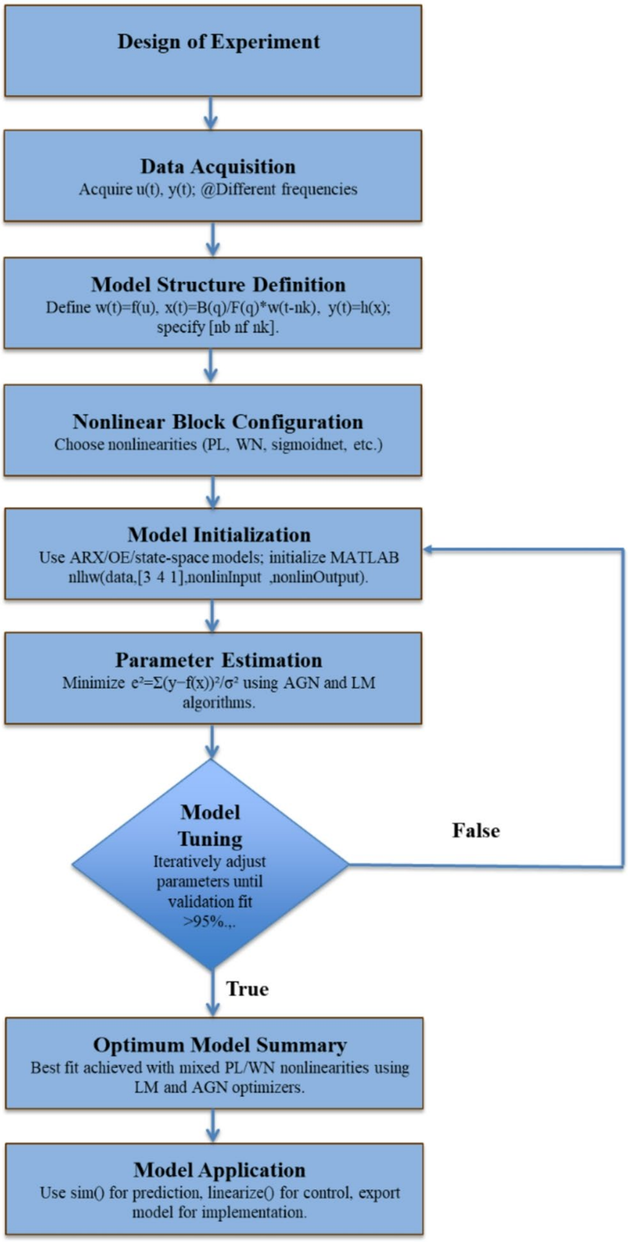


Fig. 2 Data driven modelling, model estimation, and validation of Hammerstein-Wiener system identification technique

maintained at 23 °C throughout the experiment. However, during loading of the SESMA wire, the temperature varied due to the self-heating of SESMA.

This experiment examines the frequencies of 0.01 Hz, 0.1 Hz, 1 Hz, 10 Hz, 20 Hz, and 30 Hz, with a strain amplitude ranging from 1.5% to 6%. In these circumstances, it was presumed that the displacement followed a sinusoidal wave shape.

3 Hammerstein-Wiener modelling

System identification for modelling dynamic systems mathematically using measured input and output data is shown in Fig. 2. Time-domain or frequency-domain measurements of a system's input and output signals were selected a suitable modelling framework was change based on the features of the data, and system identification is classified as either linear or nonlinear. In general, if a system operates in a linear range, it may be described using linear system identification, even if the system itself is not linear. The linear model may be insufficient for a system operating in nonlinear ranges; hence, nonlinear system identification is chosen instead. Next, the models are estimated using data from actual experiments, then the models are verified for accuracy.

Several nonlinear models, including the Autoregressive Exogenous model (NARXM), the Hammerstein model (HM), the Wiener model (WM), and the Hammerstein-Wiener model (HW), are developed for nonlinear system identification. Specifically, the Hammerstein-Wiener model provides accurate representations of nonlinear dynamic systems [28]. Hammerstein-Wiener model estimation requires uniformly sampled data in the time domain. There is one input and one output (SISO) channel for the data. The Hammerstein-Wiener structure is shown in Fig. 3, and the corresponding input-output relation is described as follows [29]:

$$w\left(t\right)=f\left(u\left(t\right)\right) \tag{1}$$

$$x\left(t\right)=\sum\nolimits_{i}^{n_u}\frac{B_i\left(q\right)}{F_i\left(q\right)}w\left(t-n_k\right) \tag{2}$$

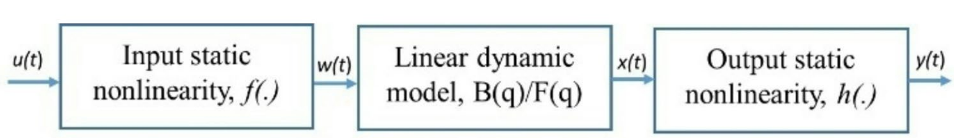
$$B\left(q\right)=b_1+b_2q^{-1}+\ldots+b_nq^{-b_u+1} \tag{3}$$

$$F\left(q\right)=1+f_1q^{-1}+\ldots+f_nq^{-f_n} \tag{4}$$

$$y\left(t\right)=h\left(x\left(t\right)\right) \tag{5}$$

Input data *u(t)* is transformed by a nonlinear function *f* in the form *w(t)* = *f(u(t))*, where *w(t)* is an internal variable that is the output of the input nonlinearity block has the same dimension as *u(t)*. Further, *x(t)*, is the output of the linear

Fig. 3 Hammerstein-Wiener system, consists of input nonlinearity followed by a static system and output nonlinearity



block, which is the ratio of two polynomials. The numerator polynomial, *B(q)* is a polynomial that has (b_n+1) zeros. The number of poles is equal to the number of coefficients in the denominator polynomials *F(q)*, which is denoted by n_f. The time-shifting operator *q*, which is equivalent to the z-transform, is found in polynomials *B* and *F* and can be expanded as in Eq. (3) and (4). An infinite number of inputs, denoted by u_n. For any given input, the time it takes for that input to be translated into an output is denoted by the number of samples, k_n, by adding up b_n and f_n, one can get the model's order. Ideally, the best model would have at least this much. It is not necessary to incorporate nonlinearity at both the input and output levels into the model structure. As discussed above, the Hammerstein-Wiener Model incorporates nonlinear functions into its nonlinear input and output blocks. Nonlinearity estimators are used in the nonlinear blocks. In this study, nonlinear blocks are employed to capture fundamental nonlinear estimators such as piecewise linear and wavelet functions. In addition to the sigmoid network's nonlinear estimators, the dead zone, saturation functions are frequently employed as estimators as well. Nonlinear functions can be estimated with the use of a wavelet (WN) estimator, which combines wavelet theory and neural networks. Using wavelets as their activation function, wavelet networks are a type of feed-forward neural network. The wavelet network function can be expressed as is the following function expansion [30]:

$$y = (u - r)\,PL + \sum_{i}^{n} as_i * f\left(bs\,(u - r)\,Q + cs\right) + \sum_{i}^{n} aw_i * g\left(bw_i\,(u - r)\,Q + cw_i\right) + d \tag{6}$$

where, functions u and y are input and output, respectively. We have a linear subspace, P, and a nonlinear subspace, Q and L is referred to as a linear coefficient. The value d represents the offset in the output. A scaling coefficient, as, and a wavelet coefficient, aw, is defined.

The coefficients bs and bw represent the dilation of the wavelet and the scaling, respectively. Coefficients of scaling, translation, and wavelet translation are denoted by cs and cw, respectively.

By employing iterative approaches in the MATLAB System Identification Toolbox, we have successfully attained a level of data fitting performance that is above the 95% criterion, even when confronted with different nonlinearity in both the input and output channels with a fixed order in the linear channel for each specific test frequency. Further details on the model and optimization procedure are presented in the next section.

4 Algorithm for model optimization

A dataset comprising measurements is considered, represented as pairs (x_i, y_i) where x_i represents the input, in this case strain, and y_i represents the observation, i.e., stress, for $i=1, 2, \ldots, N$. We posit that these measurements exhibit an underlying trend in accordance with a particular function denoted as *f(x)*. We further hypothesize that the differences between the observed values, y_i, and the values predicted by the function, $f(x_i)$, conform to a Gaussian distribution characterized by a variance denoted as σ_y^2. Employing the principle of maximum likelihood, the most favourable estimates for the parameters governing the behaviour of the function *f(x)* are those that serve to minimize the associated objective function:

$$e^2 = \sum_{i=1}^{N} \frac{\left(y_i - f(x_i)\right)^2}{\sigma_y^2} \tag{7}$$

However, when the function *f(x)* deviates from a polynomial form, the estimation of the parameters characterizing *f(x)* necessitates the application of numerical methodologies. Gauss-Newton algorithm is commonly employed to optimize parameter estimates for nonlinear models by iteratively adjusting the parameters to minimize the discrepancy between observed data and model predictions. It is especially effective when dealing with systems described by differentiable functions. On the other hand, Levenberg-Marquardt algorithm is a versatile optimization technique frequently used for nonlinear regression tasks. It combines elements of both the Gauss-Newton and gradient descent methods, offering adaptability and robustness in estimating parameters for a wide range of non-polynomial functions. The Levenberg-Marquardt (LM) algorithm is more robust in handling ill-conditioned problems, while the Adaptive Gauss-Newton method (AGN) is primarily based on the Gauss-Newton approach and may be more suitable for well-conditioned problems with good initial guesses.

4.1 Evolution of adaptive Gauss-Newton algorithm (AGN)

The Adaptive Gauss-Newton (AGN) algorithm is a widely used optimization algorithm for identifying the parameters of nonlinear models. It is an iterative procedure that modifies the model parameters based on the Hessian matrix, which provides information on the objective function's curvature [31].

The Hammerstein-Wiener model has been designed with piecewise linear nonlinearities at both the input and output

Table 1 Optimization for Hammerstein-Wiener model

Frequency (Hz)	Nonlinear		Linear	% Fit	Opt.
	I/P	O/P	[b_n f_n k_n]		
0.01	WN	WN	[3 4 1]	96.6	AGN
0.1	WN	WN	[3 4 1]	92.17	AGN
1	WN	WN	[3 4 1]	96.8	AGN
10	WN	WN	[3 4 1]	67.76	AGN
20	WN	WN	[3 4 1]	77.22	AGN
30	WN	WN	[3 4 1]	91.4	AGN
0.01	PL	PL	[3 4 1]	97.49	AGN
0.1	PL	PL	[3 4 1]	30.84	AGN
1	PL	PL	[3 4 1]	88.23	AGN
10	PL	PL	[3 4 1]	95.52	AGN
20	PL	PL	[3 4 1]	81.87	AGN
30	PL	PL	[3 4 1]	92.29	AGN

Table 2 Optimization for Hammerstein-Wiener model

Frequency (Hz)	Nonlinear		Linear	% Fit	Opt.
	I/P	O/P	[b_n f_n k_n]		
0.01	WN	WN	[3 4 1]	96.71	LM
0.1	WN	WN	[3 4 1]	92.41	LM
1	WN	WN	[3 4 1]	96.43	LM
10	WN	WN	[3 4 1]	69.16	LM
20	WN	WN	[3 4 1]	77.17	LM
30	WN	WN	[3 4 1]	96.27	LM
0.01	PL	PL	[3 4 1]	99.09	LM
0.1	PL	PL	[3 4 1]	30.84	LM
1	PL	PL	[3 4 1]	93.68	LM
10	PL	PL	[3 4 1]	68.21	LM
20	PL	PL	[3 4 1]	98.47	LM
30	PL	PL	[3 4 1]	97.23	LM

stages, complemented by a linear channel coefficient (b_n=3 f_n=4 k_n=1).

The optimization of this model is entrusted to the Adaptive Gauss-Newton algorithm; this model consistently achieves a fit of above 90% across a range of frequencies in Table 1. However, it's noteworthy that at the 10 and 20 Hz frequencies, deviations from this high-fit standard are observed. These anomalies warrant further investigation into the distinctive dynamics at play in these specific frequency domains, possibly suggesting the need for model adjustments or additional considerations in these instances. Further, employs Wavelet nonlinearities for both input and output transformations, a method well-regarded for its capacity to capture nuanced temporal dependencies and nonlinear responses [32, 33]. Complementing these nonlinearities and a selected set of same linear channel coefficients (b_n=3 f_n=4 k_n=1) maintains the integrity of the model's linear characteristics, ensuring consistent behaviour across a wide range of frequencies.

4.2 Evolution of Levenberg-Marquardt algorithm (LM)

The LM algorithm adjusts the regularization parameter dynamically based on the curvature of the objective function. When the objective function is close to being linear, the algorithm behaves like the gradient descent method. When the objective function is highly nonlinear, the algorithm behaves like the Gauss-Newton method. This adaptability makes the LM algorithm more robust and efficient compared to other optimization algorithms.

Overall, the LM algorithm is a powerful tool for identifying the parameters of the HW model for SMAs and can produce accurate predictions of the SMA behaviour under different loading conditions. However, it is important to carefully validate the results and ensure that the model accurately captures the underlying physical behaviour of the system [34].

The optimization of this model is facilitated through the Levenberg-Marquardt algorithm. The model achieves a fit, with over 90% accuracy, across various frequencies. However, it is important to acknowledge specific challenges encountered at 10 and 20 Hz frequencies, where deviations from this high-fit standard are observed. These deviations can be attributed to the unique characteristics of the hysteresis of SESMA at these frequencies, warranting further investigation and potential model refinement. In Table 2, we have presented PFE: Final prediction error and % Fit for all frequencies under test.

4.3 Combination of optimization algorithm and I/P-O/P nonlinearities

Given the observations made regarding the model's fit performance, we have undertaken an adjustment process. In response to these findings, we have modified both the input

Table 3 Optimum Hammerstein-Wiener model for SESMA

Frequency (Hz)	Nonlinear		Linear	Model properties			Opt.
	I/P	O/P	[b_n f_n k_n]	% Fit	FPE	Loss Fcn	
0.01	PL	PL	[3 4 1]	99.09	2.083	1.62	LM
0.1	PL	WN	[3 4 1]	98.4	16.9	16.11	LM
1	WN	PL	[3 4 1]	98.67	7.377	4.969	AGN
10	PL	PL	[3 4 1]	95.52	87.87	43.93	AGN
20	PL	PL	[3 4 1]	98.47	5.571	3.78	LM
30	PL	WN	[3 4 1]	98.19	25.58	9.993	LM

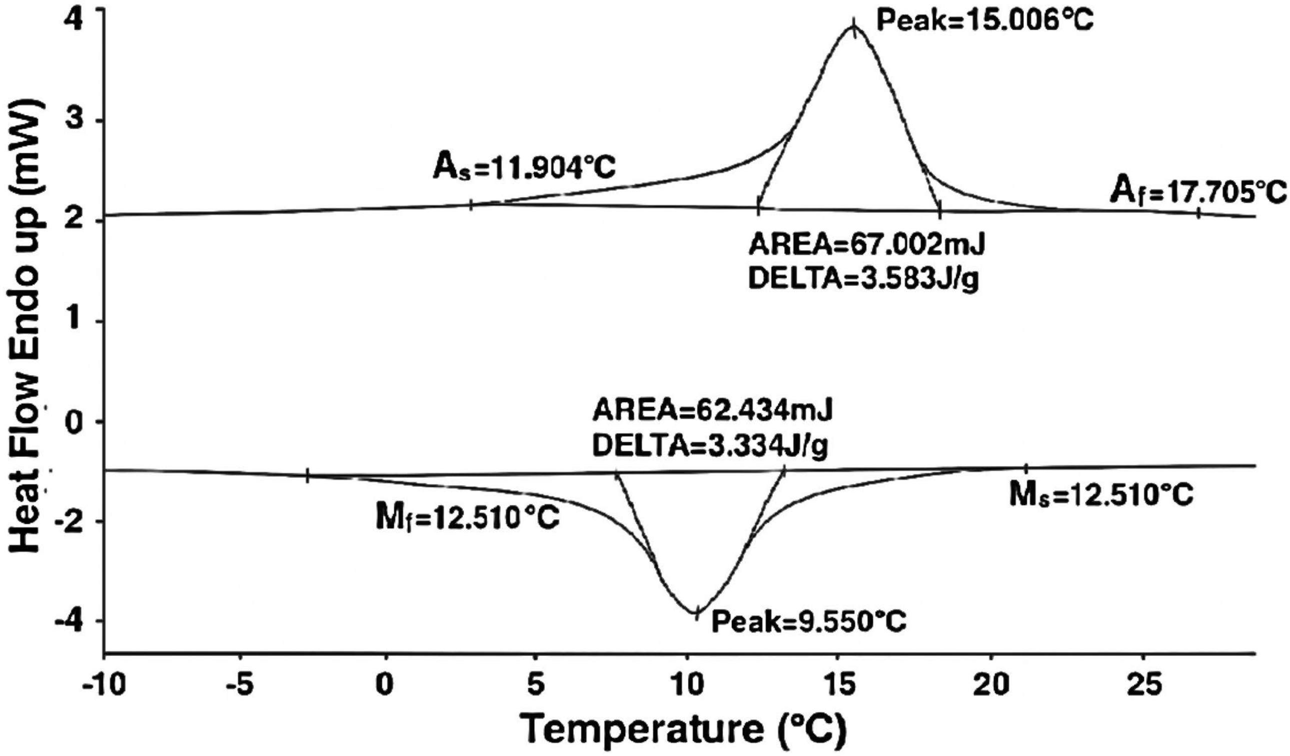


Fig. 4 DSC analysis of the SESMA wire

and output nonlinearities and have also reconfigured the optimization methodology. These adjustments have been made with the overarching objective of achieving a substantial improvement in the model's fit accuracy, aiming to surpass the 95% threshold, which is presented in Table 3.

We focus on achieving superior predictive model accuracy, exceeding 95%, through the minimization of the Final Prediction Error (FPE) and Loss Function. To attain this goal, we employed two popular optimization algorithms: The Levenberg-Marquardt (LM) algorithm and the Adaptive Gauss-Newton (AGN) algorithm. The LM algorithm combines gradient descent and Gauss-Newton methods to iteratively adjust model parameters, seeking minimal Final prediction error (FPE). The AGN algorithm dynamically approximates the Hessian matrix, adapting to cost function curvature for efficient optimization. By sequentially applying these algorithms, we optimized the model's parameters, resulting in optimal predictive accuracy.

5 Results and discussion

In this section, the discussion will encompass the differential scanning calorimetry test result prior to the tensile test, and the use of experimental data to predict SESMA cyclic behaviour through Hammerstein-Wiener (HW) models.

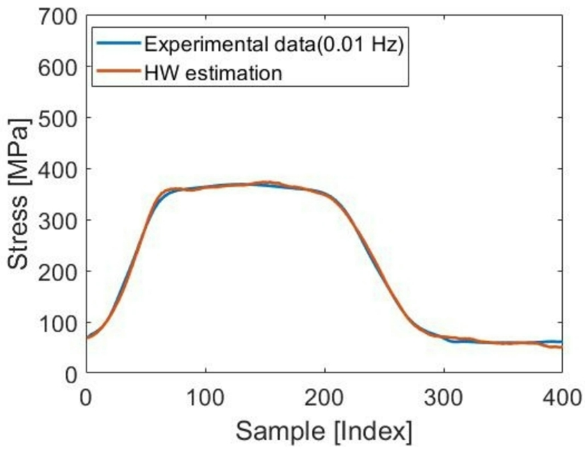
5.1 Differential scanning calorimetry result

Prior to the tensile test, the differential scanning calorimetry test is conducted to obtain the transformation temperature of the SESMA. Differential scanning calorimetry (DSC) is a thermal analysis technique that is frequently used to examine the thermal behaviour of shape memory alloys (SMA). The unique behaviour of SMAs, metallic materials that recover their original shape after deformation, is caused by a thermal or mechanical stimulus.

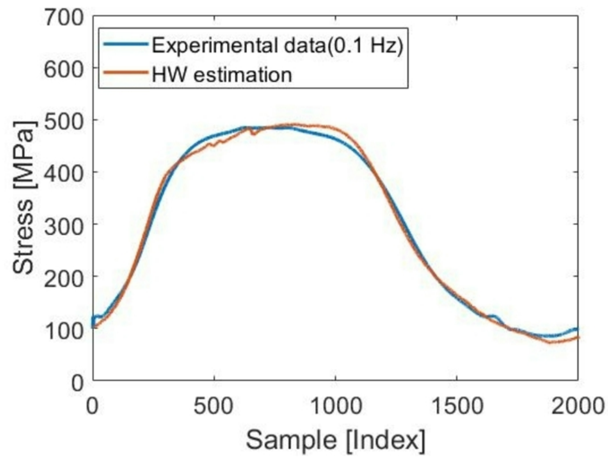
In DSC, a small sample of the SMA is heated and/or cooled, and the heat flow is measured. As the temperature changes, the DSC curve exhibits endothermic and/or exothermic peaks corresponding to phase transitions.

For instance, an endothermic peak is typically observed during heating as the SMA transitions from martensitic to austenitic phase. Conversely, an exothermic peak is observed during cooling as it transitions from austenite to martensite.

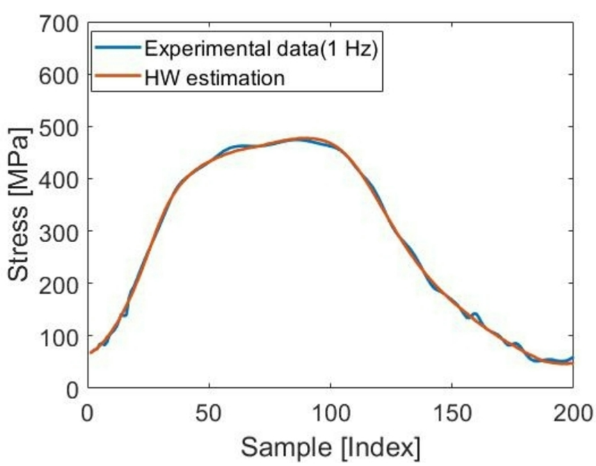
Tension-tension cycling experiments at different frequencies were performed to understand the fatigue behaviour of NiTi superelastic shape memory alloy wires (SESMA). The wire used in this investigation has a chemical composition of 54.3% Ni and 45% Ti by weight. The cold worked specimen is heat treated for 15 min at 500 °C before water quenching. The DSC result for the specimen with 0.6 mm


Experimental data(0.01 Hz)
HW estimation
Stress [MPa]
0
100
200
300
400
500
600
700
0
100
200
300
400
Sample [Index]

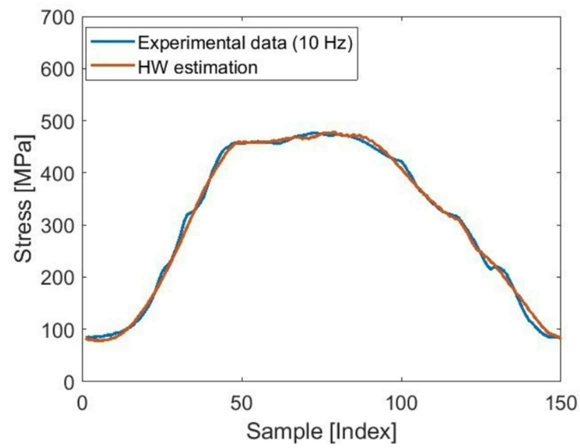
(a)


Experimental data(0.1 Hz)
HW estimation
Stress [MPa]
0
100
200
300
400
500
600
700
0
500
1000
1500
2000
Sample [Index]

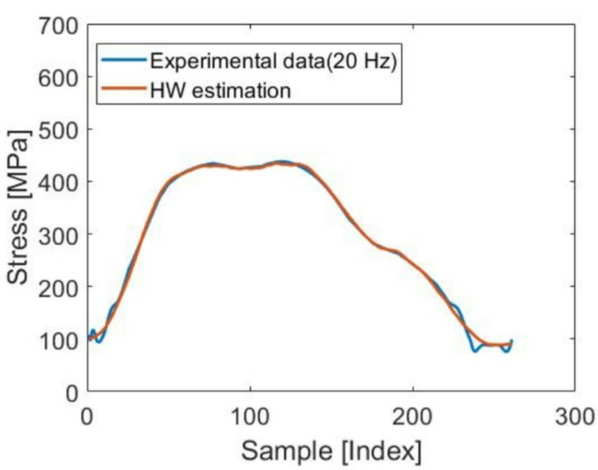
(b)


Experimental data(1 Hz)
HW estimation
Stress [MPa]
0
100
200
300
400
500
600
700
0
50
100
150
200
Sample [Index]

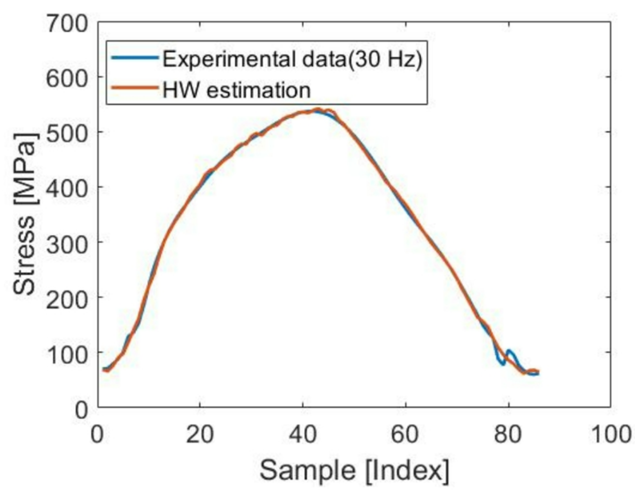
(c)


Experimental data (10 Hz)
HW estimation
Stress [MPa]
0
100
200
300
400
500
600
700
0
50
100
150
Sample [Index]

(d)


Experimental data(20 Hz)
HW estimation
Stress [MPa]
0
100
200
300
400
500
600
700
0
100
200
300
Sample [Index]

(e)


Experimental data(30 Hz)
HW estimation
Stress [MPa]
0
100
200
300
400
500
600
700
0
20
40
60
80
100
Sample [Index]

(f)

Fig. 5 Fifth cycle stress under different experimental loading frequency and corresponding Hammerstein-Wiener model validation **a** 0.01 Hz **b** 0.1 Hz **c** 1 Hz **d** 10 Hz **e** 20 Hz **f** 30 Hz

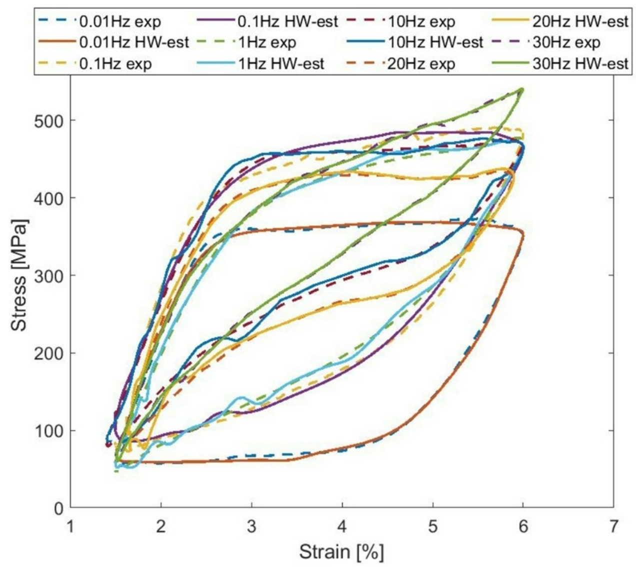


Fig. 6 Hysteresis loops under different experimental loading frequency and corresponding Hammerstein-Wiener model output

diameter is illustrated in Fig. 4. One can find transformation temperatures such as M_f (martensite finish temperature), Ms (martensite start temperature), A_f (austenite finish temperature) and As (austenite start temperature) are as follows: 6.8, 12.5, 17.7, and 11.9 °C.

5.2 Hammerstein-Wiener (HW) model result

As it was previously mentioned, SESMA wires are mechanically loaded up to a strain level of 6% from 1.5% while being maintained at a constant temperature of 23 °C. In these tests, the loading cycles at rates of 0.01, 0.1, 1, 10, 20, and 30 Hz. Using experimental data, Hammerstein-Wiener (HW) models estimate with the fifth cycle of the observed data at the tested frequencies of 0.01 Hz, 0.1 Hz, 1 Hz, 20 Hz, and 30 Hz in Fig. 5. The nonlinear Hammerstein-Wiener (NLHW) model is capable of accurately predicting SMA cyclic behaviour.

The system identification tool GUI produces a precise best fit to the data by estimating nonlinear input and output channels and modifying the linear order through iterative methods. The study focused on examining two nonlinear estimators, namely the piecewise-linear (PL) and wavelet-network estimators (WN), for inputs and outputs.

It is worth noting that the Hammerstein-Wiener (HW) model successfully calculated the frequencies evaluated, which ranged from 0.01 Hz to 30 Hz. The model's output is then compared to the experimental output in Fig. 5.

Table 4 Area of hysteresis loop corresponding to dissipated energy (Ed)

Frequency (Hz)	Experiment (MJ/m^3)	Estimation (MJ/m^3)	Error (%)
0.01	10.06	10.08	0.19
0.1	10.20	10.21	0.08
1	7.949	7.944	0.06
10	6.41	6.49	1.29
20	6.02	6.06	0.66
30	4.32	4.29	0.65

Table 5 Austenite modulus

Frequency (Hz)	Experiment (GPa)	Estimation (GPa)	Error (%)
0.01	30.39	30.61	0.70
0.1	28.66	28.81	0.51
1	27.24	27.55	1.13
10	30.39	30.48	0.29
20	34.48	34.75	0.77
30	30.22	29.90	1.07

Although the proposed model may estimate any cycle using the available experimental data, we used the fifth cycle to ensure the accuracy of the depiction.

The hysteresis (energy dissipation) loop is an essential aspect of SESMA. The region enclosed by the stress-strain hysteresis loop signifies the energy dissipated per unit volume during the cycle. The hysteresis loop using a second cycle compared with experimental data using above mention frequencies in Fig. 6. The region of the hysteresis represents the heat-based energy loss throughout that test cycle. The frequencies of 0.01 Hz, 0.1 Hz, 1 Hz, 10 Hz, 20 Hz, and 30 Hz are taken into consideration for a strain amplitude of 1.5–6.5%. Table 4 presents the total area of the hysteresis loop corresponding to dissipated energy (E_d) for the experimental second cycle and the fifth cycle for estimation conditions. As one moves up the frequency of the test, one notices a shrinking of the area occupied by those frequencies. Moreover, there is a good correlation between the model and experimental values, with a maximum error of 1.29%.

Larger hysteresis loops have been observed to correspond with greater energy dissipation. At low frequencies, like 0.01 Hz, the phase transformation occurs more completely, allowing for greater energy dissipation than at 30 Hz.

The strain caused by the cross-head movement is accounted for when determining the austenite modulus. In Table 5, we have calculated the experimental and estimated austenite modulus. One can note that there is only a little change in austenite's modulus with frequency over the range of interest. Our experimental and estimated values for the austenite modulus are within the reported 20–50 GPa range [35] in the literature.

6 Conclusion

In this study, the experimental stress and strain data obtained from SESMA at cycle frequencies of 0.01 Hz, 0.1 Hz, 1 Hz, 10 Hz, 20 Hz, and 30 Hz were used to construct the Hammerstein-Winner model for estimating cyclic behaviour. The nonlinearity of the input and output channels, however, varies with frequency. The linear channel of the model coefficient ($bn=3$, $fn=4$, $kn=1$) remained the same throughout all frequencies that were examined. We validated the fifth cycle, but the model is flexible enough to estimate any cycles (stress-strain) on any data from experiments. At each frequency, the stress estimation provided by the model was close to the experimental results. From the model output, we were also able to calculate the austenite modulus as well as the stress-strain hysteresis area. These results showed a strong correlation, with an error that was no greater than 1.89%, when compared to the results of the experiments.

In conclusion, the study represents a state-of-the-art/attractive method for model prediction of cyclic behaviour; it is limited by its superelastic behaviours and lack of consideration for thermal effects and rate-dependent phase transformations. Future study will focus on improving the model by integrating thermomechanical coupling and adaptive components, as well as applying it to real-world scenarios to evaluate its practical applicability in fatigue life estimation.

Acknowledgements The authors wish to thank the Director, CSIR-National Aerospace Laboratories, Bengaluru and Head, ALD, CSIR-NAL, for their support. The authors would like to extend their acknowledgment to Dr. V.L. Sateesh, ACD, CSIR-NAL, for the useful discussions while analysing the data. Mr. Arunabha Datta would like to thank the Principal, Ramgarh Engineering College, for providing support during his Ph.D work.

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